Past Paper 2024 Reconduct.

(1) Prove that for a hermition operator the eigen vectors corresponding to different eigen value are orthogonal,

We known that as the given eigen function is

$$\hat{A}\psi_1 = a_1\psi_1 \quad \text{——} (1)$$
$$\hat{A}\psi_2 = a_2\psi_2 \quad \text{——} (2)$$

Taking conjugate - (1) & (2)

$$A^*\psi_1^* = a_1\psi^* \quad \text{——} (3)$$
$$\hat{A}^*\psi_1^* = a^*\psi_2^* \quad \text{——} (4)$$

As the hermition operator is

$$\int_{-\infty}^{\infty}\psi^*\hat{A}\psi\,dt = \int_{-\infty}^{\infty}\hat{A}\,\psi^*\psi\,dt$$

$$\int_{-\infty}^{\infty}\psi^* a_2\psi_2\,dt = \int_{-\infty}^{\infty} a_1\psi_2^*\psi\,dt$$

$$a_2\int_{-\infty}^{\infty}\psi^*\psi\,dt = a_1\int_{-\infty}^{\infty}\psi_2^*\psi\,dt$$

$$a_2\int_{-\infty}^{\infty}\psi^*\psi\,dt - a_1\int_{-\infty}^{\infty}\psi^*\psi\,dt = 0$$

$$(a_2 - a_1)\int_{-\infty}^{\infty}\psi^*\psi\,dt = 0$$

$$a_2 - a_1 \neq 0 \qquad \text{So } \int_{-\infty}^{\infty}\psi^*\psi\,dt = 0$$

function is orthogonal.

(2) What is correspondence Principle,

The correspondence principle is state that the Quantum number must be agree in classical theory in the limit of large Quantum number are known as correspondence principle.

$$E_n = \frac{13.6e}{n^2}$$

③ Prove that $[L_x, L^2] = 0$

We know that

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

so

$$[L_x, L^2] = [L_x, L_x^2 + L_y^2 + L_z^2]$$

$$[L_x, L^2] = [L_x, L_x^2] + [L_x, L_y^2] + [L_x + L_z^2]$$

$$[L_x, L^2] = [L_x, L_x L_x] + [L_x, L_y, L_y] + [L_x, L_z L_z]$$

$$[L_x, L^2] = \cancel{[L_x, L_x] L_x} + \cancel{[L_x, L_x] L_x} + [L_x, L_y] L_y + [L_x, L_y] L_y + [L_x, L_z] L_z + [L_x, L_z] L_z$$

$$[L_x, L^2] = [L_x, L_y] L_y + [L_x, L_y] L_y + [L_x, L_z] L_z + [L_x, L_z] L_z$$

$$// = \cancel{i\hbar L_z L_y} + \cancel{i\hbar L_z L_y} - \cancel{i\hbar L_y L_z} - \cancel{i\hbar L_y L_z}$$

$[L_x, L^2] = 0$ Hence prove.

④ Differnce b/w bounded system and Unboundes System.

Bounded system

When a partical are move in system they can't across to system on potential wall because the partical has less energy due to Poteantial wall. That's why they can't away for the system
→ It spectrum will be discrete.

Unbounded System.

When a partical are cross to the system on ~~due to~~ potential wall due to hyer energy of partical this system are called unbounded system.
It spectrum will be contineous.

⑥ Define unitary operator:. A linear operator $\hat{U}$ is said to be unitary if its inverse $\hat{U}^{-1}$ is equal to its adjoint ie U^{+}

$$U^{+} = U^{-1} \quad \text{or} \quad UU^{+} = U^{+}U = I$$

(6) Write any two Postulate of Quantum Mechanics?

→ A quantum system is a completly described by a wave function which contain all information about system. Its denoted by ψ

→ Every Measurable Physical Quantity is represented hermition operator like as Position Momentum energy

(7) Prove that

Expectation value of hermition operator is real

Define:

$$\int_{-\infty}^{\infty} \psi^* A \psi \, dt = \int_{-\infty}^{\infty} \hat{A} \psi^* \psi \, dt$$ ⇐ Hermition operator

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \psi^* |\hat{x}| \psi \, dt$$ ⇐ expectation value

Solve

$$\langle \hat{A} \rangle = \int_{-\infty}^{\infty} \psi^*(x) A \psi(x) \, dx$$

Taking conjugate

$$\langle \hat{A} \rangle^* = \int_{-\infty}^{\infty} \left(\psi^*_{(x)} \hat{A} \psi(x)\right)^* dx$$

$$\langle \hat{A} \rangle^* = \int_{-\infty}^{\infty} (\hat{A}\psi)^* \psi \, dx$$

use hermition operator is real

$$\int_{-\infty}^{\infty} (A\psi)^* \psi \, dx = \int \psi^* \hat{A} \psi \, dx$$

so

$$\langle \hat{A} \rangle^* = \langle \hat{A} \rangle$$

Here Prove the Expectation value of hermition operator is real.

→ Expectation value of anti hermition operator is imaginary.

Solved

We known that expectation value is given that

$$\langle \hat{A} \rangle = \int_{-\infty}^{\infty} \psi^*_{(x)} (\hat{A}) \psi_{(x)} dx \quad \text{--- (1)}$$ Conjugate

$$\langle \hat{A} \rangle^* = \int_{-\infty}^{\infty} (\psi^*_{(x)} (\hat{A}) \psi_{(x)})^* dx \quad \text{--- (2)}$$

$$\langle \hat{A} \rangle^* = \int_{-\infty}^{\infty} (A\psi)^* \psi \, dx$$ Antihermition operator

$$\hat{A}^\dagger = -\hat{A}$$

$$\downarrow$$

$$-\langle \hat{A} \rangle$$

So

$$\int_{-\infty}^{\infty} (\hat{A}\psi)^* \psi \, dx = \int_{-\infty}^{\infty} \psi^* (-\hat{A}) \psi \, dx$$

$$\int_{-\infty}^{\infty} (\hat{A}\psi)^* \psi \, dx = -\int_{-\infty}^{\infty} \psi^* A \psi \, dx$$

$$\langle \hat{A}^\dagger \rangle = -\langle \hat{A} \rangle$$ hence proved.

(8) Write an expression for identity operator and completeness relation for dirac Notation

Dirac Notation.

Dirac Notation is a Quantum system to represented by a hilbert Space. So the Quantum state like as ket ($|\psi\rangle$) and bra ($\langle\psi|$) Symbols.

Identity operator:

An operator which Not change to original state are known Identity operator.

Completeness Relation

$$\sum_n |n\rangle\langle n| = \hat{I}$$

(9) Can we determine z-component of angular momentum and azimuthal angle with Precision Simultaneously?

اچھا جیسا ہی azimuthal آتا ہے تو اس کو ہم ϕ سے ظاہر کرتے ہیں —

$\phi \rightarrow$ azimuthal.
$L_z \rightarrow$ angle component

$[L_z, \phi]$

We known that

$[L_z, \phi]\psi = [L_z\phi - \phi L_z]\psi$

$= \left[-i\hbar \frac{\partial}{\partial\phi} \phi\psi - \phi(-i\hbar)\frac{\partial\psi}{\partial\phi}\right]$

Apply differential.

$= \left[-i\hbar\psi - i\hbar\phi\frac{\partial\psi}{\partial\phi} + i\hbar\phi\frac{\partial\psi}{\partial\phi}\right] =$

$[L_z, \phi]\psi = -i\hbar\psi$

$[L_z\, \phi] = -i\hbar$

Similarly

$[\phi, L_z] = [i\hbar]$

Hence Proved.

سوال میں دو چیزوں کے بارے میں پوچھا گیا ہے تو اس میں ہم اگر z کو ~~بڑھائے تو~~ بہت اچھے سے پیمائش کریں تو ہم پھر ϕ کو ناپ نہیں سکتے اور اگر ہم ϕ کو ناپے تو z کو معلوم نہیں کر سکتے کیونکہ یہ Uncertainty Principle کے خلاف ہے تو ہم دونوں چیزوں کو ایک ساتھ ناپ نہیں سکتے۔

$\Delta\phi\,\Delta L_z \geq \hbar/2$

چونکہ ان دونوں کے درمیان nonzero ہے۔

(10) Prove the following properties of Dirac Notation

i) Schwarz inequality

In any two Quantum state $|\psi\rangle$ and $|\phi\rangle$ of hilbert space we can show that in Dirac Notation is

$|\langle\phi|\psi\rangle|^2 \leq \langle\phi|\phi\rangle . \langle\psi|\psi\rangle$

and eigen function

$|\psi\rangle = \alpha|\phi\rangle$

α is a scalar quantity

→ This inequality has a linearly independent vector Q.S

یہ صرف دو vector کا magnitude ان کی inner product کی limit کو بتاتا ہے ان کا product ہمیں زیادہ نہیں ہو سکتا یہ صرف cosine angle کی طرح ہے

ii) Triangle inequality.

Triangle inequality gives a bound on the norm of a sum of vector.

In Dirac Notation

$|\psi\rangle + |\phi\rangle \leq |\psi\rangle + |\phi\rangle$

$\sqrt{\langle\psi+\phi|\psi+\phi\rangle} \leq \sqrt{\langle\psi|\psi\rangle} + \sqrt{\langle\phi|\phi\rangle}$

یہ دو vector کے مجموعہ کرنے کے بعد جو total آتی ہے وہ ہمیشہ ان دونوں کی لمبائی کے مجموعے سے کم یا برابر ہوتی ہیں۔

(11) What is the condition for which the sum of two projection operator $\hat{p}_1$ and $\hat{p}_2$ is also a projection operator.

In given condition when the two projection operator are sum each other but the cannot return as orignal condition, so they are Not a projection operator

(12) Statment of principle of superposition –

When two or more wave moving through the same region of space waves will superimpose and produce a will defined combined effect. If $\psi = a_1\psi_1 + a_2\psi_2$ is a Solution here a_1, a_2 = constant. Then ψ_1 & ψ_2 fellow the superposition Principle that means the net amplitude causes by two or more wave travelly same space is sum of amplitude that would have been produced by individual wave seperately:

$$\hat{O}\psi = \lambda\psi$$

(13) Write down the expression for the momentum operator in the position representation

The momentum operator in the position representation is given by

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}$$

It acts on the wave function $\psi(x)$ as

$$\hat{p}\psi(x) = -i\hbar\frac{\partial}{dx}\psi(x)$$

(14) Write down the equation of continuity showing the conservation of probability in Q.M?

A

یہ یاد رکھنا ہے کہ اس کا جواب سمجھ ہو

آئے گا اور ہم اس کو ثابت کرنے کے لیے Schrodinger

time dependent eq کا استعمال کریں گے ۔

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = i\hbar\frac{\partial\psi}{\partial t} \quad \text{---} \ (1)$$

complex conjugate

$$-\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^* = -i\hbar\frac{\partial\psi^*}{\partial t} \quad \text{---} \ (2)$$

xly by eq 1 in ψ^* & eq 2 in ψ with both are substract

$$-\frac{\hbar^2}{2m}\left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right) = i\hbar\left(\psi^*\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi^*}{\partial t}\right)$$

~~xly i~~ $\div \hbar$ on b.s.

$$-\frac{\hbar}{2m}\left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right) = i\left(\psi^*\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi^*}{\partial t}\right)$$

Divided by i on b.s

$$-\frac{\hbar}{2im}\left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right) = \left(\psi^*\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi^*}{\partial t}\right)$$

$$-\frac{\hbar}{2im}\left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right) = \frac{\partial}{\partial t}\left(\psi^*\psi\right)$$

$$\frac{\partial}{\partial t}\psi^*\psi + \frac{\hbar}{2im}\nabla\left(\psi^*\nabla\psi - \psi\nabla\psi^*\right) = 0$$

Compare by analyze

$$\frac{\partial \rho}{\partial t}(\vec{r}, t) + \nabla \cdot \vec{J} = 0$$

$\rho(\vec{r}, t) \rightarrow$ probability density

$\vec{J} \rightarrow$ Current density.

(15) Write down the expression for time rate of change of expectation value of an operator A,

The time rate of change of expectation value of an operator $\hat{A}$ is given by

$$\frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle + \left\langle\frac{\partial A}{\partial t}\right\rangle$$

اگر C.M کی condition کی صورت میں دیکھا جائے تو اس میں ہم $\langle \hat{A} \rangle = \int \psi A \psi^* dx$ کو expectation value لکھتے ہیں۔ اور ψ فنکشن ہوتا ہے t اور x کا لیکن جب ہم اس کی پیمائش کرتے ہیں تو اس وقت اس کو d/dt سے ضرب دیتے ہیں۔

$$\frac{d}{dt}\langle \hat{A} \rangle = \frac{d}{dt}\int \psi^*(x,t)\,\hat{A}\,\psi(x,t)\,dx$$

اور تین چیزوں پر depend کرتا ہے۔ پھر ہم Schrodinger سے اس کو حل کرتے ہیں اور جواب

$$\frac{d}{dt}[A] = \frac{i}{\hbar}\int \psi^*[\hat{H}\hat{A} - \hat{A}\hat{H}]\psi\, dx + \int \psi\,\frac{\partial A}{\partial t}\,\psi\, dx$$

$$\frac{d}{dt}\langle \hat{A} \rangle = \frac{1}{i\hbar}\langle [\hat{A}, \hat{H}] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle$$

Long Question

$$P.E = \frac{qV}{4\epsilon_0 \pi r} \Rightarrow \frac{-e^2}{4\pi\epsilon_0 r}.$$

Central Potential:

A central Potential is a type of P.E that depends only on the distance from a central point (usually the origin.

$$V(\vec{r}) = V(r)$$

Solution of Radial eq/ -

We know that radial eq/ as given that (M)

$$\left[\frac{1}{r^2}\frac{d}{dr}\left[r^2\frac{dR}{dr}\right]+\left[\frac{2m}{\hbar^2}\left(E+\frac{e^2}{4\pi\epsilon_0 r}\right)-\frac{l(l+1)}{r^2}\right]R=0\right.$$

$$\left[\frac{1}{r^2}\frac{d}{dr}\left[r^2\frac{dR}{dr}\right]+\left[\frac{2mE}{\hbar^2}+\frac{2me^2}{4\pi\epsilon_0 r\hbar^2}-\frac{l(l+1)}{r^2}\right]R=0\right.$$

$$\frac{1}{r^2}\left(r^2\frac{d^2R}{dr^2}+2r\frac{dR}{dr}\right)+\left[\frac{2mE}{\hbar^2}+\frac{2}{r}\left(\frac{me^2}{4\pi\epsilon_0\hbar^2}\right)-\frac{l(l+1)}{r^2}\right]R=0$$

$$\frac{d^2R}{dr^2}+2r\frac{dR}{dr}+\left[\frac{2mE}{\hbar^2}+\frac{2}{r}\left(\frac{me^2}{4\pi\epsilon_0\hbar^2}\right)-\frac{l(l+1)}{r^2}\right]R=0 \quad \text{(23)}$$

let

$$-\alpha^2=\frac{2mE}{\hbar^2} \quad \text{— 24(a)}$$

$$\lambda=\frac{me^2}{4\pi\epsilon_0\hbar^2\alpha} \quad \text{— (24b)} \qquad \lambda\alpha=\frac{me^2}{4\pi\epsilon_0\hbar^2} \quad \text{— (24c)}$$

24a, ~~24b~~, 24c put in 23

$$\frac{d^2R}{dr^2}+\frac{2}{r}\frac{dR}{dr}+\left[-\alpha^2+\frac{2\lambda\alpha}{r}-\frac{l(l+1)}{r^2}\right]R=0 \quad \text{(25)}$$

let

$$\rho=2\alpha r$$

Apply deriv—

$$\boxed{\frac{d\rho}{dr}=2\alpha} \qquad \boxed{r=\frac{\rho}{2\alpha}} \quad \text{— (26a)}$$

$$\frac{dR}{dr}=\frac{dR}{d\rho}\frac{d\rho}{dr} \Rightarrow \boxed{\frac{dR}{dr}=\frac{dR}{d\rho}2\alpha} \quad \text{— (26b)}$$

again derive w.r.t r

$$\frac{d^2R}{dr^2}=\frac{d^2R}{d\rho^2}\cdot\left(\frac{d\rho}{dr}\right)^2 .$$

کیونکہ ρ خود ایک r کا فنکشن ہے

$$\boxed{\frac{d^2R}{dr^2}=4\alpha^2\frac{d^2R}{d\rho^2}} \quad \text{— (26c)}$$

Putting 26a, 26b, 26c in 25

$$4\alpha^2 \frac{d^2R}{d\rho^2} + \frac{4\alpha}{\rho}\left(2\alpha \frac{dR}{d\rho}\right) + \left[-\alpha^2 + \frac{4\lambda\alpha^2}{\rho} - \frac{l(l+1)}{(\rho/2\alpha)^2}\right]$$

$$4\alpha^2 \frac{d^2R}{d\rho^2} + \frac{8\alpha^2}{\rho}\frac{dR}{d\rho} + \left[-\alpha^2 + \frac{4\alpha^2\lambda}{\rho} - \frac{4\alpha^2}{\rho^2} l(l+1)\right] R = 0$$

$$\frac{d^2R}{d\rho^2} + \frac{2}{\rho}\frac{dR}{d\rho} + \left[\frac{-1}{4} + \frac{\lambda}{\rho} - \frac{1}{\rho}(l(l+1))\right] R = 0 \quad \text{— (27)}$$

اگر ρ کو ہم ∞ کر دیں تو ادھر سے باقی صرف $-1/4$ رہ جائے گا

$\rho \to \infty$

$$\boxed{\frac{d^2R}{d\rho^2} - \frac{R}{4} = 0} \quad \text{— (28)}$$

Similarly apply root eq

$R(\rho) = N e^{-\rho/2}$ where N is Normalization form.

The exact solution of 27 is given by

$$\frac{d^2R}{d\rho^2} + \frac{2}{\rho}\frac{dR}{d\rho} + \left[\frac{-1}{4} + \frac{\lambda}{\rho} - \frac{l(l+1)}{\rho^2}\right] = R.$$

we known that R is function of ρ and $F(\rho)$

$$R(\rho) = f(\rho) N e^{-\rho/2}$$

$$R = N e^{-\rho/2} \cdot f \quad \text{— (29a)}$$

Apply derivative

$$\frac{dR}{d\rho} = N\left[e^{-\rho/2}\frac{df}{d\rho} + e^{-\rho/2}\left(\frac{-1}{2}\right) f\right]$$

$$\frac{dR}{d\rho} = N\left[e^{-\rho/2}\frac{df}{d\rho} - \frac{e^{-\rho/2}}{2} f\right] \quad \text{— (29b)}$$

again der

$$\frac{d^2R}{d\rho^2} = N\left[e^{-\rho/2}\frac{d^2f}{d\rho^2} + e^{-\rho/2}\frac{-1}{2}\frac{df}{d\rho}\right] - \frac{1}{2}\left[e^{-\rho/2}\frac{df}{d\rho} + e^{-\rho/2}\cdot\frac{-1}{2} f\right]$$

$$\frac{d^2R}{d\rho^2} = N\left[e^{-\rho/2}\frac{d^2f}{d\rho^2} \boxed{- \frac{1}{2}e^{-\rho/2}\frac{df}{d\rho} - \frac{1}{2}e^{-\rho/2}\frac{df}{d\rho}} + \frac{1}{4}e^{-\rho/2} f\right]$$

-1

$$\frac{d^2R}{d\rho^2} = N\left[e^{-\rho/2}\frac{d^2f}{d\rho^2} - e^{-\rho/2}\frac{df}{d\rho} + \frac{1}{4}e^{-\rho/2} f\right] \quad \text{— (29c)}$$

Putting value 29a, 29b, 29c in 27

$$N\left[e^{-\rho/2}\frac{d^2f}{d\rho^2} - e^{-\rho/2}\frac{df}{d\rho} + \frac{1}{4}e^{-\rho/2}f\right] + \frac{2N}{\rho}\left[e^{-\rho/2}\frac{df}{d\rho} - \frac{1}{2}e^{-\rho/2}f\right]$$

$$+\left[\frac{-1}{4} + \frac{\lambda}{\rho} - \frac{l(l+1)}{\rho^2}\right]Ne^{-\rho/2}f = 0$$

Divided $Ne^{-\rho/2}$

$$\frac{d^2f}{d\rho^2} - \frac{df}{d\rho} + \frac{1}{4}f + \frac{2}{\rho}\left[\frac{df}{d\rho} - \frac{1}{2}f\right] +$$

$$\left[\frac{1}{4} + \frac{\lambda}{\rho} - \frac{l(l+1)}{\rho^2}\right]f = 0$$

$$\frac{d^2f}{d\rho^2} - \frac{df}{d\rho} + \cancel{\frac{1}{4}f} + \frac{2}{\rho}\frac{df}{d\rho} - \frac{f}{\rho} - \cancel{\frac{f}{4}} + \frac{f\lambda}{\rho} - \frac{l(l+1)f}{\rho^2} = 0$$

$$\frac{d^2f}{d\rho^2} + \left(\frac{2}{\rho} - 1\right)\frac{df}{d\rho} + \left[\frac{\lambda}{\rho} - \frac{1}{\rho} - \frac{l(l+1)}{\rho^2}\right]f = 0$$

find value.

$$\frac{d^2f}{d\rho^2} + \left(\frac{2-\rho}{\rho}\right)\frac{df}{d\rho} + \left[\frac{(\lambda-1)}{\rho} - \frac{l(l+1)}{\rho^2}\right]f = 0 \quad \text{(30)}$$

let

$$f(\rho) = \rho^l G(\rho) \quad \text{(31a)}$$

$$\frac{df}{d\rho} = \left(\rho^l\frac{dG}{d\rho} + l\rho^{l-1}G\right) \quad \text{(31b)}$$

Apply U.V rule

$$\frac{d^2f}{d\rho^2} = \left(\rho^l\frac{d^2G}{d\rho^2} + l\rho^{l-1}\frac{dG}{d\rho}\right) + \left(l\rho^{l-1}\frac{dG}{d\rho} + l(l-1)\rho^{l-2}G\right)$$

$$\frac{d^2f}{d\rho^2} = \rho^l\frac{d^2G}{d\rho^2} + l\rho^{l-1}\frac{dG}{d\rho} + l\rho^{l-1}\frac{dG}{d\rho} + l(l-1)\rho^{l-2}G = 0 \quad \text{(31c)}$$

$$\left[\rho^l\frac{d^2G}{d\rho^2} + 2l\rho^{l-1}\frac{dG}{d\rho} + l(l-1)\rho^{l-2}G\right] + \left(\frac{2}{\rho} - 1\right)$$

$$\left[\rho^l\frac{dG}{d\rho} + l\rho^{l-1}G\right] + \left[\frac{(\lambda-1)}{\rho} - \frac{l(l+1)}{\rho^2}\right]\rho^l G = 0$$

$$\rho^l \longrightarrow \rho^{l-1}$$

$$\left[\rho^1\rho^{l-1}\frac{d^2G}{d\rho^2} + l\rho^{l-1}G + 2l\rho^{l-1}\frac{dG}{d\rho} + l(l-1)\rho^{l-1}\rho^{-1}G\right] +$$

$$\left(\frac{2}{\rho} - 1\right)\left[\rho^1\rho^{l-1}\frac{dG}{d\rho} + l\rho^{l-1}G\right] + \left[\frac{(\lambda-1)}{\rho} - \frac{l(l+1)}{\rho^2}\right]\rho^1\rho^{l-1}G = 0$$

$\div \rho^{l-1}$ on b.s.

$$\left[\rho \frac{d^2G}{d\rho^2} + 2l\frac{dG}{d\rho} + l(l-1)\frac{G}{\rho}\right] + \left(\frac{2}{\rho}-1\right)\left[\rho\frac{dG}{d\rho} + lG\right]$$
$$+ \left[\frac{\lambda-1}{\rho} - \frac{l(l+1)}{\rho^2}\right]\rho G = 0$$

ρ common $\Downarrow \div$

$$\left[\rho \frac{d^2G}{d\rho^2} + 2l\frac{dG}{d\rho} + l(l-1)\frac{G}{\rho}\right] + \left(\frac{2}{\rho}-1\right)\left[\rho\frac{dG}{d\rho} + lG\right] +$$
$$\left[(\lambda-1) - \frac{l(l+1)}{\rho}\right] G = 0$$

$\frac{dG}{d\rho}$ comm

$$\rho\frac{d^2G}{d\rho^2} + \frac{dG}{d\rho}\left[2l + \left(\frac{2}{\rho}-1\right)\rho\right] + G\left[\frac{l(l-1)}{\rho} + \left(\frac{2}{\rho}-1\right)l + \lambda - 1\right.$$
$$\left. - \frac{l(l+1)}{\rho}\right] = 0$$

$$\rho\frac{d^2G}{d\rho^2} + \frac{dG}{d\rho}\left[2l+2-\rho\right] + G\left[\frac{l^2}{\rho} - \frac{l}{\rho} + \frac{2l}{\rho} - l + \lambda - 1 - \frac{l^2}{\rho} - \frac{l}{\rho}\right] = 0$$

$$\rho\frac{d^2G}{d\rho^2} + \left[2l+1+1-\rho\right]\frac{dG}{d\rho} + \left[(\lambda-1-l) + \frac{1}{\rho}\cancel{(l^2 - l + 2l - l^2 - l)}\right]G = 0$$

$$\rho\frac{d^2G}{d\rho^2} + \left[(2l+1)+1-\rho\right]\frac{dG}{d\rho} + (\lambda-1-l)G = 0$$

let $\lambda = n$

$$\rho\frac{d^2G}{d\rho^2} + \left[(2l+1)+1-\rho\right]\frac{dG}{d\rho} + (n-1-l)G = 0$$

let

$$2l+1 = p \qquad n+l = n'$$

then

$$n'-p = n+l-p = n+l-(2l+1)$$
$$n'-p = n+l-2l-1$$

$$\boxed{\rho\frac{d^2G}{d\rho^2} + \left[p+1-\rho\right]\frac{dG}{d\rho} + (n'-p)G = 0} \quad (32)$$

This eq shows that laguerre's differential eq is given as

$$x\frac{d^2}{dx^2}L_n^m(x) + (m+1-x)\frac{d}{dx}L_n^m(x) + (n-m)L_n^m(x) = 0 \quad (33)$$

comparing (32) & (33)

The solution of eq^n (33) give the expression for $L_n^m(x)$ are called associated laguerres polynomials.
Thus the solution of eq^n (32) is given as.

$$G(s) = L_n^m(x) \Rightarrow G(p) = L_{n'}^{p}(s)$$

$$\boxed{G(s) = L_{n+1}^{2l+1}(s)} \quad \text{—— (34)}$$

using 34 in 31 a we get

$$\boxed{F(s) = s^l L_{n+1}^{2l+1}(s)} \quad \text{—— (35)}$$

Put 35 in 29 a

$$\boxed{R(s) = Ne^{-s/2} s^l L_{n+1}^{2l+1}(s)} \quad \text{—— (36)}$$

ہمیں یاد ہے کہ جو فنکشن تھا وہ ρ میں R(r) کے
صورت میں دیا گیا تھا اس نے ہم ρ کو r
میں تبدیل کریں گے

$$s = 2\alpha r \quad \text{—— (a)} \qquad \lambda\alpha = \frac{me^2}{4\pi\epsilon_0\hbar^2} \Rightarrow \alpha = \frac{me^2}{4\pi\epsilon_0\hbar^2\lambda}$$

$$s = \frac{2r}{\lambda}\left(\frac{me^2}{4\pi\epsilon_0\hbar^2}\right)$$

$$s = \frac{2r}{n}\left(\frac{me^2}{4\pi\epsilon_0\hbar^2}\right) \qquad \therefore \boxed{a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}} \text{ is known as bohr's orbit}$$

$$s = \frac{2r}{n}\frac{1}{a_0} \Rightarrow s = \frac{r2}{na_0} \quad \text{—— (37)}$$

using 37 in 36 we get

$$R(r) = Ne^{-s/2}\left(\frac{2r}{na_0}\right)^l L_{n+1}^{2l+1}\left(\frac{2r}{na_0}\right)$$

$$R(r) = Ne^{-\left(\frac{2r}{na_0}\right)\frac{1}{2}}\left(\frac{2r}{na_0}\right)^l L_{n+1}^{2l+1}\left(\frac{2r}{na_0}\right)$$

$$R(r) = Ne^{-\frac{r}{na_0}}\left(\frac{2r}{na_0}\right)^l L_{n+1}^{2l+1}\left(\frac{2r}{na_0}\right) \quad \text{—— (38)}$$

Normalize condition. $\int R^*(r) R(r) r^2 dr = 1$

$R^{\#}_{r}$ = ہیں i کیونکہ ہوگی same کی دقسم ہے

$$N = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n\{(n+l)!\}^3}} \quad \text{—} (39)$$

Putting 39 in 38

$$R(r) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-(l+1))!}{2n\{(n+l)!\}^3}} \; e^{-r/na_0} \left(\frac{2r}{na_0}\right)^l L^{2l+1}_{n+l}\left(\frac{2r}{na_0}\right)$$

This is called radial equation..

اگر ہم n کو l+1 سے چھوٹا فرض کریں تو مساوی ساوی

ignore ہو جائے گی

The neccessary condition to obtain the above solution is that $n \geq l+1$

n is principle Quantum number.

$n = 1, 2, 3 \ldots \ldots l+1$

$$\alpha^2 = \frac{-2mE}{\hbar^2}$$

$$\left(\frac{me^2}{4\pi\varepsilon_0 \hbar^2 \lambda}\right)^2 = \frac{-2mE}{\hbar^2}$$

$$E = \frac{-\hbar^2}{2m} \frac{m^2 e^4}{(4\pi\varepsilon_0)^2 \hbar^4 n^2} \qquad \lambda = n$$

$$\boxed{E_n = -\frac{13.6}{n^2}\,eV}$$

which is in accordance with Bohrs Model of H-atom.

General

Uncertainty relation b/w two operator.

There are two different hermite operator as given below.

$\hat{A} = (\hat{A}^\dagger) \qquad \hat{B} = (\hat{B}^\dagger)$

$\langle \Delta A \rangle = \langle \hat{A} \rangle - \langle \hat{A} \rangle \qquad \langle \Delta B \rangle = \langle \hat{B} \rangle - \langle B \rangle$

$\langle (\Delta \hat{A})^2 \rangle = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \qquad \langle (\Delta B)^2 \rangle = \langle \hat{B}^2 \rangle - \langle \hat{B} \rangle^2$

let

$$\langle \psi | \psi \rangle \geq 0$$

$$\langle \phi | (\Delta \hat{A} - i\lambda \Delta \hat{B})(\Delta \hat{A} + i\lambda \Delta \hat{B}) | \phi \rangle \geq 0$$

$$\langle \phi | (\Delta A)^2 + i\lambda \Delta \hat{A} \Delta B - i\lambda \Delta \hat{B} \Delta A - i^2 \lambda^2 (\Delta \hat{B})^2 | \phi \rangle \geq 0$$

$$\langle \phi | (\Delta \hat{A})^2 + i\lambda [\hat{A}\hat{B}] + \lambda^2 (\Delta \hat{B})^2 | \phi \rangle \geq 0 \quad \text{—— (1)}$$

apply differential w.r.t

$$\frac{d}{d\lambda}\langle\phi | \cancel{(\Delta \hat{A})^2}^{\,0} + i\lambda [\hat{A} B] + \lambda^2 (\Delta \hat{B})^2 | \phi \rangle = 0$$

$i^2 = -1$
$(-i^2) = (-1, i)^2$
$(-1)^2 (i^2)(1 \cdot -1) = -1$

$$\frac{d}{d\lambda} \langle \phi | i\lambda [\hat{A}\hat{B}] + \lambda^2 (\Delta \hat{B})^2 | \phi \rangle = 0$$

$$\langle \phi | i [\hat{A}\hat{B}] | \phi \rangle + 2\lambda (\Delta \hat{B})^2 = 0$$

$$\lambda = \frac{-i \langle \phi | [\hat{A}\hat{B}] | \phi \rangle}{2 \langle (\Delta \hat{B})^2 \rangle} \quad \text{Putting value } \lambda \text{ in (1)}$$

$$\langle (\Delta \hat{A})^2 \rangle + \frac{|\langle \phi | [\hat{A}\hat{B}] | \phi \rangle|^2}{2 \langle (\Delta \hat{B})^2 \rangle} + \left(\frac{-i \langle \phi | [\hat{A}\hat{B}] | \phi \rangle}{2 \langle (\Delta \hat{B})^2 \rangle} \right)^2$$

$$\langle (\Delta \hat{A})^2 \rangle + \frac{|\langle \phi | [\hat{A}\hat{B}] | \phi \rangle|^2}{2 \langle (\Delta \hat{B})^2 \rangle} - \frac{|\langle \phi | [\hat{A}\hat{B}] | \phi \rangle|^2 \cancel{(\Delta \hat{B})^2}}{4 \langle (\Delta \hat{B})^4 \rangle}$$

$$\langle (\Delta \hat{A})^2 \rangle + \frac{[\langle \phi | [\hat{A}\hat{B}] | \phi \rangle]^2}{2 \langle (\Delta \hat{B})^2 \rangle} - \frac{|\langle \phi | [\hat{A}\hat{B}] | \phi \rangle|^2}{4 \langle (\Delta \hat{B})^2 \rangle}$$

$$\langle (\Delta A)^2 \rangle + \frac{|\langle \phi [\hat{A}\hat{B}] | \phi \rangle|^2}{4 \langle (\Delta B)^2 \rangle} \geq 0$$

$$\langle (\Delta A)^2 \rangle \geq - \frac{|\langle \phi | \hat{A}\hat{B} | \phi \rangle|^2}{4 \langle (\Delta B)^2 \rangle}$$

$$\langle (\Delta A)^2 \rangle \langle (\Delta B)^2 \rangle \geq - \frac{|\langle \phi | [\hat{A}\hat{B}] | \phi \rangle|^2}{4}$$

$\cancel{\langle (\Delta A)^2 \rangle \langle (\Delta B)^2 \rangle} \geq$

let $[\hat{A}\hat{B}] = i\hat{C}$

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle = -\frac{|\langle\phi|i\hat{C}|\phi\rangle|^2}{4}$$

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle = -\frac{i^2}{4}|\langle\phi|\hat{C}|\phi\rangle|^2$$

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle = \frac{1}{4}|\langle\phi|\hat{C}|\phi\rangle|^2$$ Taking s.r b.s

$$\langle\Delta\hat{A}\Delta B\rangle = \frac{1}{2}\langle\hat{C}\rangle$$ hence proved.

Q3

Find the eigen function and correspondence eigenvalue for a 1 dimensional harmonic oscillator using the algebraic method and prove the energy spectrum both discrete of degeneration

June 17, 2025

1 Introduction

This document mathematically proves that the energy spectrum of the 1-dimensional quantum harmonic oscillator (1D QHO) is both discrete and non-degenerate. We will utilize the algebraic method, employing ladder operators, which offers an elegant solution to this fundamental problem in quantum mechanics.

2 Hamiltonian of the Harmonic Oscillator

The Hamiltonian $\hat{H}$ for a 1D QHO is given by:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where $\hat{x}$ is the position operator, $\hat{p}$ is the momentum operator, m is the mass of the particle, and ω is the angular frequency of the oscillator.

The fundamental commutation relation between position and momentum operators is:

$$[\hat{x}, \hat{p}] = i\hbar$$

2.1 Definition of Ladder Operators

We define the annihilation operator ($\hat{a}$) and the creation operator ($\hat{a}^\dagger$) as:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)$$

These operators satisfy the commutation relation:

$$\left[\hat{a}, \hat{a}^\dagger\right] = 1$$

Let's prove this commutation relation:

$$
\begin{aligned}
[\hat{a}, \hat{a}^\dagger] &= \left[\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right), \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)\right] \\
&= \frac{m\omega}{2\hbar}\left[\hat{x} + \frac{i}{m\omega}\hat{p}, \hat{x} - \frac{i}{m\omega}\hat{p}\right] \\
&= \frac{m\omega}{2\hbar}\left([\hat{x}, \hat{x}] - \left[\hat{x}, \frac{i}{m\omega}\hat{p}\right] + \left[\frac{i}{m\omega}\hat{p}, \hat{x}\right] - \left[\frac{i}{m\omega}\hat{p}, \frac{i}{m\omega}\hat{p}\right]\right) \\
&= \frac{m\omega}{2\hbar}\left(0 - \frac{i}{m\omega}[\hat{x}, \hat{p}] + \frac{i}{m\omega}[\hat{p}, \hat{x}] - 0\right) \\
&= \frac{m\omega}{2\hbar}\left(-\frac{i}{m\omega}(i\hbar) + \frac{i}{m\omega}(-i\hbar)\right) \\
&= \frac{m\omega}{2\hbar}\left(\frac{\hbar}{m\omega} + \frac{\hbar}{m\omega}\right) \\
&= \frac{m\omega}{2\hbar}\left(\frac{2\hbar}{m\omega}\right) \\
&= 1
\end{aligned}
$$

3 Hamiltonian in Terms of Ladder Operators

Let's express the Hamiltonian in terms of $\hat{a}$ and $\hat{a}^\dagger$. Consider the product $\hat{a}^\dagger\hat{a}$:

$$
\begin{aligned}
\hat{a}^\dagger\hat{a} &= \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right) \\
&= \frac{m\omega}{2\hbar}\left(\hat{x}^2 + \frac{i}{m\omega}\hat{x}\hat{p} - \frac{i}{m\omega}\hat{p}\hat{x} - \frac{i^2}{m^2\omega^2}\hat{p}^2\right) \\
&= \frac{m\omega}{2\hbar}\left(\hat{x}^2 + \frac{i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x}) + \frac{1}{m^2\omega^2}\hat{p}^2\right) \\
&= \frac{m\omega}{2\hbar}\left(\hat{x}^2 + \frac{i}{m\omega}[\hat{x}, \hat{p}] + \frac{1}{m^2\omega^2}\hat{p}^2\right)
\end{aligned}
$$

Substitute $[\hat{x}, \hat{p}] = i\hbar$:

$$
\begin{aligned}
\hat{a}^\dagger\hat{a} &= \frac{m\omega}{2\hbar}\left(\hat{x}^2 + \frac{i}{m\omega}(i\hbar) + \frac{1}{m^2\omega^2}\hat{p}^2\right) \\
&= \frac{m\omega}{2\hbar}\left(\hat{x}^2 - \frac{\hbar}{m\omega} + \frac{1}{m^2\omega^2}\hat{p}^2\right) \\
&= \frac{m\omega}{2\hbar}\hat{x}^2 - \frac{1}{2} + \frac{1}{2m\hbar\omega}\hat{p}^2
\end{aligned}
$$

Now, rearrange this to connect with the Hamiltonian:

$$\begin{aligned}
\hat{a}^\dagger \hat{a} + \frac{1}{2} &= \frac{m\omega}{2\hbar}\hat{x}^2 + \frac{1}{2m\hbar\omega}\hat{p}^2 \\
\hbar\omega\left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) &= \hbar\omega\left(\frac{m\omega}{2\hbar}\hat{x}^2 + \frac{1}{2m\hbar\omega}\hat{p}^2\right) \\
&= \frac{1}{2}m\omega^2\hat{x}^2 + \frac{1}{2m}\hat{p}^2
\end{aligned}$$

This is exactly the Hamiltonian $\hat{H}$. Thus, we can write:

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right)$$

We define the **number operator** $\hat{N}$ as:

$$\hat{N} = \hat{a}^\dagger \hat{a}$$

So the Hamiltonian becomes:

$$\hat{H} = \hbar\omega\left(\hat{N} + \frac{1}{2}\right)$$

4 Finding the Energy Spectrum (Eigenvalues)

Let's find the eigenstates of $\hat{N}$. Assume there exists an eigenstate $|n\rangle$ such that:

$$\hat{N}|n\rangle = n|n\rangle$$

where n is the eigenvalue. Applying the Hamiltonian to this eigenstate:

$$\hat{H}|n\rangle = \hbar\omega\left(\hat{N} + \frac{1}{2}\right)|n\rangle$$

$$\hat{H}|n\rangle = \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle$$

So the energy eigenvalues are:

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right)$$

4.1 Proof of Discreteness

To show that n must be a non-negative integer, we use the properties of the ladder operators. First, consider the expectation value of the number operator:

$$\left\langle \hat{N} \right\rangle = \langle\psi|\hat{a}^\dagger \hat{a}|\psi\rangle = |\hat{a}|\psi\rangle\rangle\langle\hat{a}|\psi\rangle| = \|\hat{a}|\psi\rangle\|^2 \geq 0$$

Since the norm squared of any state must be non-negative, the eigenvalues n of $\hat{N}$ must also be non-negative ($n \geq 0$).

Next, consider the action of the ladder operators on an eigenstate $|n\rangle$:

$$\begin{aligned}
\hat{N}(\hat{a}\,|n\rangle) &= \hat{a}^{\dagger}\hat{a}\hat{a}\,|n\rangle \\
&= (\hat{a}\hat{a}^{\dagger} - 1)\hat{a}\,|n\rangle \quad (\text{since } \hat{a}^{\dagger}\hat{a} = \hat{a}\hat{a}^{\dagger} - 1) \\
&= \hat{a}(\hat{a}^{\dagger}\hat{a} - 1)\,|n\rangle \\
&= \hat{a}(\hat{N} - 1)\,|n\rangle \\
&= \hat{a}(n - 1)\,|n\rangle \\
&= (n - 1)(\hat{a}\,|n\rangle)
\end{aligned}$$

This shows that if $|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue n, then $\hat{a}\,|n\rangle$ is an eigenstate with eigenvalue $n-1$. The operator $\hat{a}$ is thus an "annihilation" or "lowering" operator.

Similarly for $\hat{a}^{\dagger}$:

$$\begin{aligned}
\hat{N}(\hat{a}^{\dagger}\,|n\rangle) &= \hat{a}^{\dagger}\hat{a}\hat{a}^{\dagger}\,|n\rangle \\
&= \hat{a}^{\dagger}(\hat{a}^{\dagger}\hat{a} + 1)\,|n\rangle \quad (\text{since } \hat{a}\hat{a}^{\dagger} = \hat{a}^{\dagger}\hat{a} + 1) \\
&= \hat{a}^{\dagger}(\hat{N} + 1)\,|n\rangle \\
&= \hat{a}^{\dagger}(n + 1)\,|n\rangle \\
&= (n + 1)(\hat{a}^{\dagger}\,|n\rangle)
\end{aligned}$$

This shows that $\hat{a}^{\dagger}\,|n\rangle$ is an eigenstate with eigenvalue $n+1$. The operator $\hat{a}^{\dagger}$ is a "creation" or "raising" operator.

Since $n \geq 0$, there must be a lowest possible eigenvalue. Let's assume this lowest state is $|0\rangle$, for which $\hat{a}$ acting on it yields zero (it cannot go lower):

$$\hat{a}\,|0\rangle = 0$$

Applying the number operator to this state:

$$\hat{N}\,|0\rangle = \hat{a}^{\dagger}\hat{a}\,|0\rangle = \hat{a}^{\dagger}(0) = 0$$

So, the lowest possible integer value for n is $\$0.Byrepeatedlyapplying^{\dagger}$ to the ground state $|0\rangle$, we can generate all higher energy states:

$$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^{\dagger})^{n}\,|0\rangle$$

This confirms that n must take only non-negative integer values: $n = 0, 1, 2, 3, \ldots$. Therefore, the energy values $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$ are discrete. The spacing between consecutive energy levels is constant:

$$E_{n+1} - E_n = \hbar\omega\left((n+1) + \frac{1}{2}\right) - \hbar\omega\left(n + \frac{1}{2}\right) = \hbar\omega$$

5 Proving Non-Degeneracy

Degeneracy means that two or more distinct quantum states have the same energy eigenvalue. In the 1D QHO, each energy level E_n is uniquely determined by the quantum number n. The ladder operator method uniquely constructs each eigenstate $|n\rangle$ from the ground state $|0\rangle$ by applying the creation operator $\hat{a}^\dagger$ n times. For a given value of n, there is only one corresponding state $|n\rangle$. There are no other independent quantum numbers in the 1D system that could lead to different states having the same energy. For example, in 3D systems, angular momentum quantum numbers can introduce degeneracy. However, in 1D, no such additional quantum numbers exist. Since each energy level E_n corresponds to only one unique quantum state $|n\rangle$, there is no degeneracy in the 1D quantum harmonic oscillator.

6 Conclusion

The energy spectrum of the 1D quantum harmonic oscillator is mathematically proven to be:

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, 3, \ldots$$

- **Discrete:** The quantum number n takes only non-negative integer values, leading to quantized energy levels with a constant spacing of $\hbar\omega$.
- **Non-degenerate:** Each energy level E_n corresponds to a unique quantum state $|n\rangle$, as there are no additional quantum numbers in one dimension to cause multiple states to have the same energy.

harmonic oscillator are shown in fig 3.3.

Wave functions:-If we apply lowering operator $\hat{a}$ to lowest possible state ψ_0, we get zero *i.e.*

in harmonic oscillator wave function is equal to eigen function so can be called as eigen function for SHO

$$\hat{a}\psi_0 = 0 \quad \Rightarrow \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})\psi_0 = 0 \quad \Rightarrow m\omega\hat{x}\psi_0 + i\left(-i\hbar\frac{d}{dx}\right)\psi_0 = 0$$

$$\Rightarrow m\omega x\psi_0 = -\hbar\frac{d\psi_0}{dx} \quad \Rightarrow \frac{d\psi_0}{\psi_0} = -\frac{m\omega}{\hbar}xdx \quad \Rightarrow ln\psi_0 = -\frac{m\omega x^2}{2\hbar} + lnA_0$$

$$\Rightarrow ln\psi_0 = lne^{-\frac{m\omega x^2}{2\hbar}} + lnA_0 \quad \Rightarrow ln\psi_0 = lnA_0e^{-\frac{m\omega x^2}{2\hbar}} \quad \Rightarrow \psi_0 = A_0e^{-\frac{m\omega x^2}{2\hbar}}$$

Put $\frac{m\omega}{\hbar} = \alpha$, then

$$\psi_0 = A_0e^{-\alpha x^2/2}$$

By applying normalization condition, we can find A_0. So normalized wave function for ground state of harmonic oscillator is,

$$\psi_0 = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

This is a brief theory of harmonic oscillator...